SPCKI Digital Monitoring Platform: Complete Breakdown Report

Combines the **Terms of Reference [ToR]**, the **Technical Document [TD]** and the **additions from our design work ★**. Corrected items are marked ✎. Stack: PERN. Deadline and approval workflow are out of scope.

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1. Background and Objectives

The Solar Powered Community Kitchen Initiative (SPCKI) is a managed clean-cooking service: solar PV + hybrid inverter + battery + backup generator + electric appliances, run as a shared community kitchen (**16 households per kitchen ★**). The pilot must give evidence on **usage, reliability, household experience, and readiness for continuation or scale-up**.

The platform replaces paper and spreadsheets with: **Field Data Collection → Central Database → Automatic KPI Calculation → Dashboard → Reports and Decisions.**

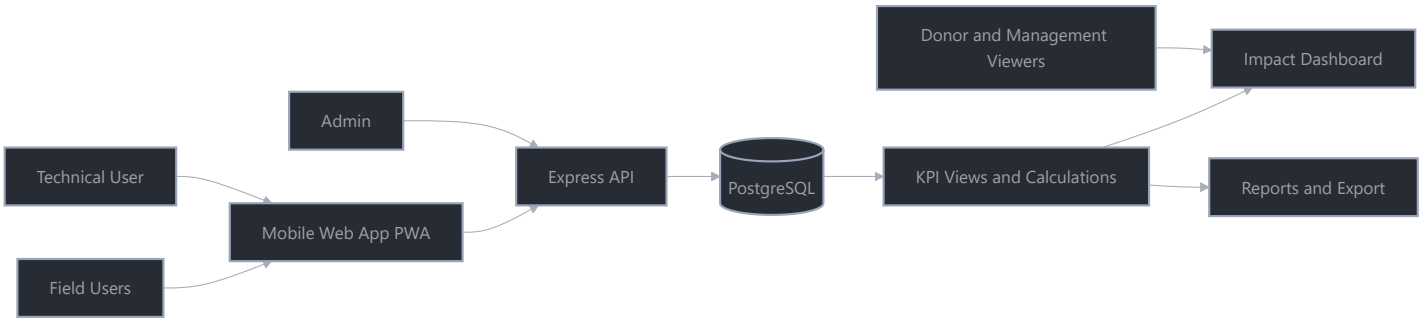
Objectives [ToR]: simplify field data entry; improve consistency; cut manual calculation; show operational and technical performance in time; monitor household experience; support donor reporting; allow data export; build evidence for scale-up. It must stay a **focused, practical tool, not a complex MIS**.

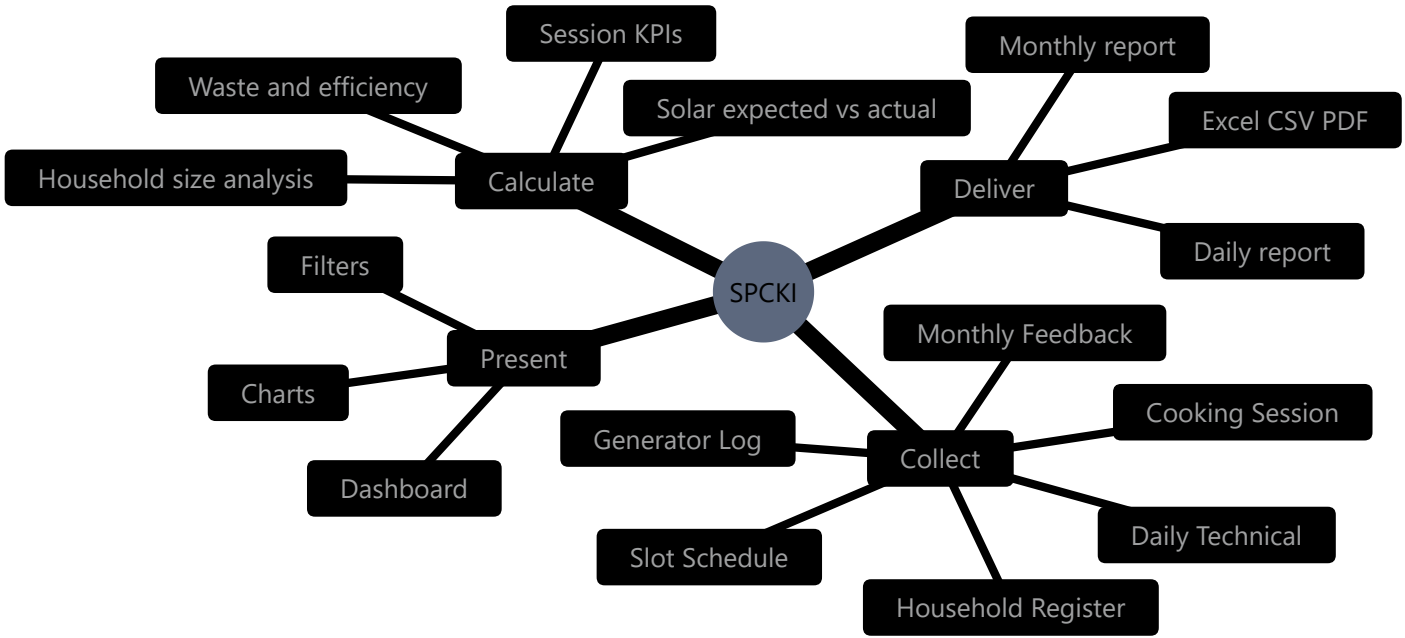
2. Scope

Four components working as one system [ToR]: **(1)** mobile-responsive data collection app, **(2)** central backend + database, **(3)** impact dashboard, **(4)** reporting and export.

Nine modules [TD]: User Management, Cooking Session Data, Daily Technical Data, Generator Maintenance, Monthly Household Feedback, KPI & Analytics, Impact Dashboard, Reports & Export, Graphic Presentation. ★ Added: Household Register, Slot Scheduling, Energy Analysis, Offline Sync.

3. System Context and Data Flow





4. Users, Roles and Permissions

Categories [ToR]: Field/Data Entry, Technical, Project/Admin, Dashboard/View-Only.

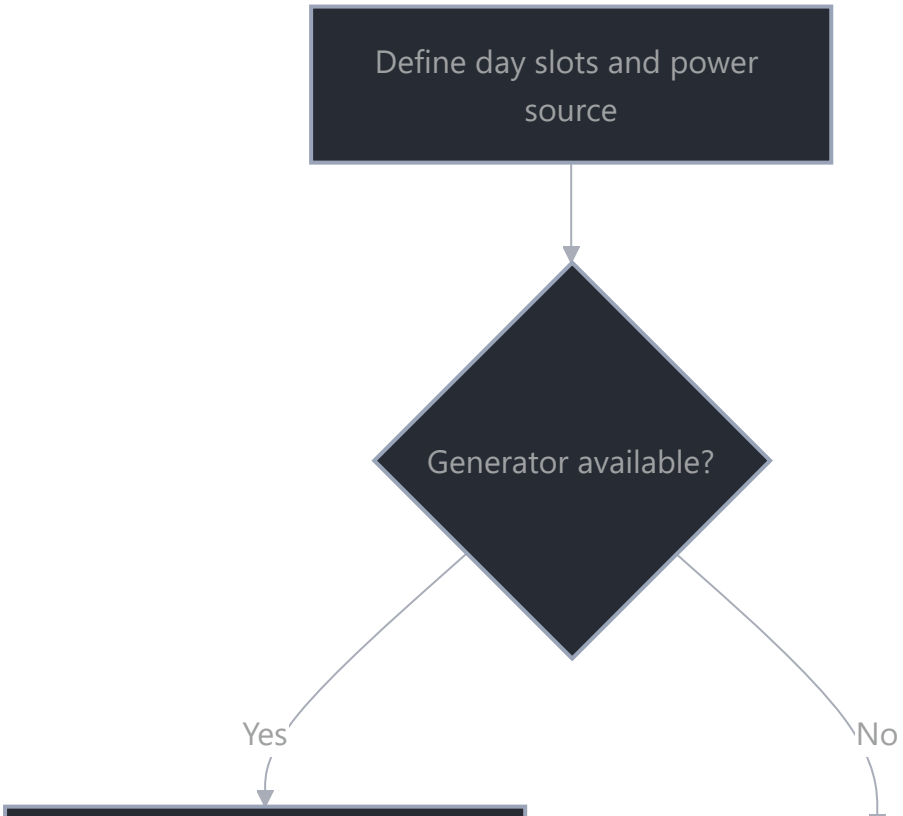
Role	Enters data	Edits / corrects	Manages users & households	Dashboard	Export
Field / Data Entry	Cooking sessions, feedback	Own entries	No	Limited	No
Technical	Daily technical, generator	Own entries	No	Technical view	No
Admin	All	All, with reason logged	Yes	Full	Yes
Viewer	No	No	No	Full, coded IDs only	Summary

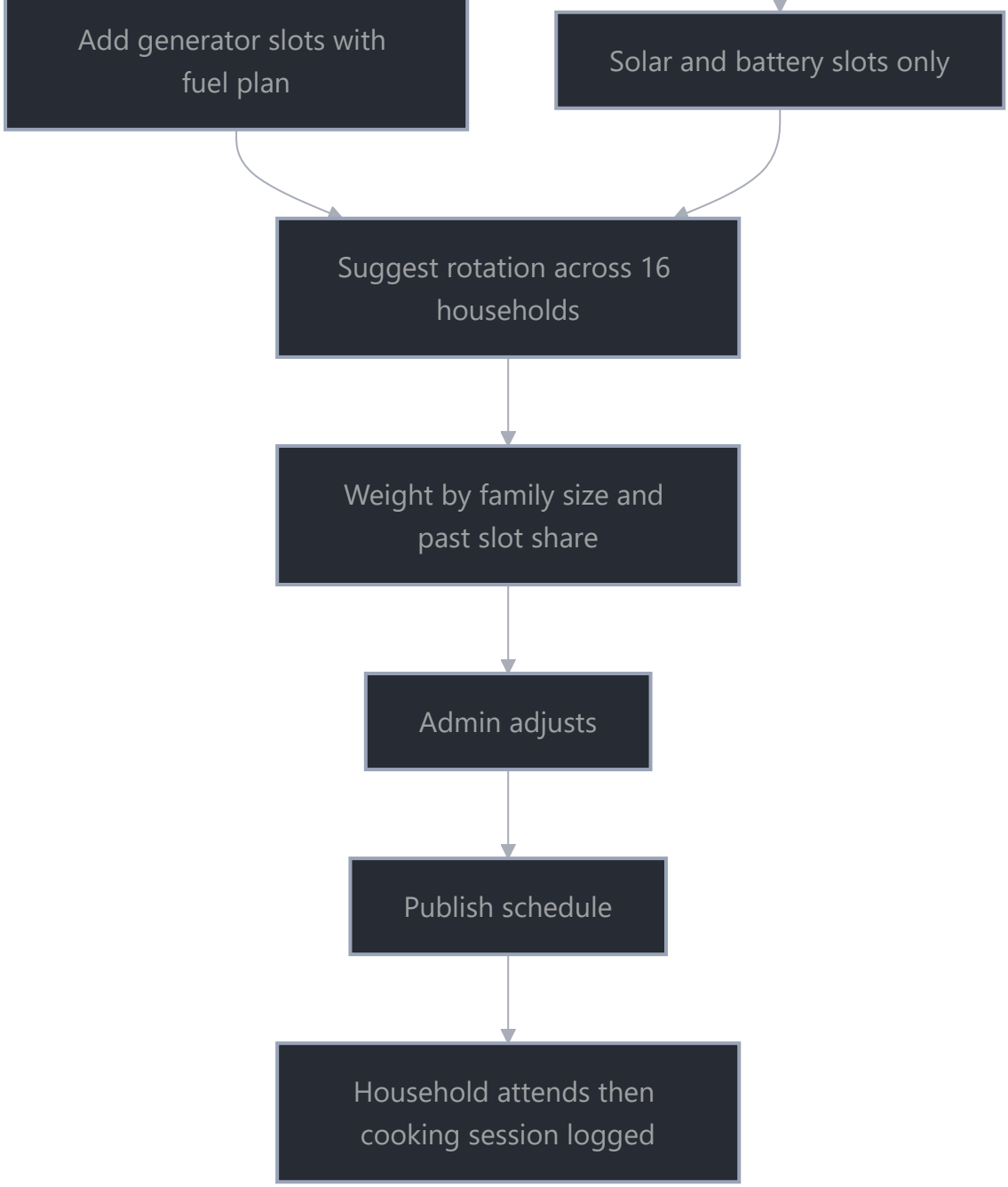
User management features [TD]: account create / update / deactivate, RBAC, permission management, secure login and password management, kitchen-wise user control (✎ replaces "department/branch-wise"), activity and audit trail, login history, user-wise and role-wise reports.



5. Operating Model: Households, Slots, Generator ★

- 16 households per kitchen (configurable). Each day has **slots** with start/end time and expected **power source: Solar / Battery / Generator**.
- Slots depend on available kitchen time and generator backup. The system **suggests a fair rotation** (equal share of peak and generator slots, optional longer slots for larger families); the admin can override.
- Slot status: `scheduled → attended / missed / cancelled / swapped`.





6. Data Dictionary (all TD tables, corrected and extended)

Legend: **TD** = in technical document, **ToR** = ToR only, ★ added, ✎ corrected.

6.1 Household Register ★

Field	Type	Note
household_code	Coded ID	No names on dashboards
kitchen_id	Ref	
status	Active/Inactive	
family_size, children_count, adults_count	Number	Updated by monthly survey
baseline_fuel_type, baseline_monthly_fuel_cost	Dropdown / amount	Needed for "baseline vs current" [ToR]
registered_at	Date	

6.2 Slot Schedule ★

Field	Type

kitchen_id, date, slot_no	Ref / date / number
start_time, end_time	Time
expected_power_source	Solar / Battery / Generator
household_id, status	Ref / scheduled, attended, missed, cancelled, swapped

6.3 Cooking Session (TD §3)

Field	Description	Source
Date	Cooking date, auto	TD
Household ID	Participating household, from slot ✎ one household per row	TD/ToR
Session	Session number, linked to slot ✎	TD
Persons cooked for	Defaults to family size	★
Start Time / End Time	Cooking start / end	TD
Cooking Duration	**Auto-calculated**	TD
Food Type	Food category ✎ multi-select (rice, dal, vegetables, fish, meat, curry, khichuri, other)	TD/ToR
Appliance	Appliance used (each has power_kw)	TD/★
Power source in use	Solar / Battery / Generator	★
Cooking Completed	Yes/No	TD
Waiting Issue	Yes/No ✎ + reason category	TD/★
Technical Problem	Yes/No ✎ + problem category	TD/★
Cleaning Completed	Yes/No	TD
Remarks	Additional information	TD

Automatic: duration, session status, relevant KPIs.

6.4 Daily Technical Data (TD §4)

Field	Description	Source
Date	Monitoring date, one per kitchen per day	TD
Daily Solar Production	kWh	TD
Morning / Evening Battery SOC	% (0-100)	TD
System Operational	Yes/No	TD
Technical Faults	Number	TD
Total Downtime	Hours	TD
Safety Incident	Yes/No ✎ + severity and note	TD/★

Weather Condition	Category (Sunny, Partly cloudy, Cloudy, Rainy, Stormy)	TD
Energy Consumed	kWh	★
Remarks	Technical notes	TD

6.5 Generator Maintenance (TD §5)

Field	Description	Source
Date	Operation date	TD
Fuel Feed ✎	Fuel in tank at start (L)	TD
Re-Fuel Feed	Litres added	TD
Fuel at end	Litres	★
Running Time	Hours	TD
Maintenance Required	Yes/No	TD
Maintenance Type	Category	TD
Remarks	Additional information	TD

Fuel consumed = start + refuel − end.

6.6 Monthly Household Feedback (TD §6)

Indicator	Response	Source
Backup Fuel Use	Yes/No	TD
Backup Fuel Expenditure	Amount	TD
Perceived reduction in backup fuel	1-5	ToR ★
Safety, Privacy, Waiting Experience, Overall Satisfaction, Kitchen Cleanliness, Ease of Cooking	1-5 each (1 = Very Poor, 5 = Very Good)	TD
Slot time suitability	1-5	★
Time enough for family size	Too short / About right / Too long	★
Current family members	Number	★
Willingness to Continue	Yes/No	TD
Remarks	Optional	TD

7. KPIs and Analytics

7.1 Automatic KPIs (TD §8, with additions)

KPI	Calculation	Source
Participation Rate	Active HH ÷ Registered HH × 100 (✎ active = ≥1 attended session in period)	TD

Session Completion Rate	Completed ÷ Total sessions × 100	TD
Average Cooking Time	Total cooking time ÷ Completed sessions	TD
Cleaning Compliance	Cleaned ÷ Total × 100	TD
Waiting Issue Rate	Waiting sessions ÷ Total × 100	TD
Technical Fault Rate	Fault sessions ÷ Total × 100 (daily fault count kept separately) ✎	TD
System Uptime	(Scheduled − downtime) ÷ Scheduled × 100 ✎ scheduled hours from config	TD
Average Downtime	Total downtime ÷ Operational days	TD
Average Satisfaction	Total rating ÷ Responses	TD
Minutes per person	Duration ÷ persons cooked for	★
Slot utilisation	Attended ÷ Scheduled × 100	★
Backup-fuel reduction	(Baseline − current) ÷ Baseline × 100	★
Generator fuel consumed	Start + refuel − end	★

7.2 Household Size vs Cooking Time ★

Raw duration is misleading (a 5-child family may take 2 hours for rice while a 3-member family takes longer). Reports therefore show: average duration and minutes per person by **size band (1-3, 4-5, 6-7, 8+)**; duration by dish × size band; outliers (> 1.5 × median for same dish and band) with their power source, appliance and problem reason; "time enough?" feedback by size band. This separates **family size effects from power or appliance problems**.

7.3 Solar: Expected, Efficiency, Waste ★

- Expected kWh = kWp × peak sun hours × weather factor × performance ratio (0.70-0.80). Starting factors (editable, calibrate with data): Sunny 1.0, Partly cloudy 0.65, Cloudy 0.35, Rainy 0.20, Stormy 0.10.
- Efficiency % = actual ÷ expected × 100.
- Unused energy = generated − consumed − (Δ SOC ÷ 100 × battery kWh).
- Downtime loss = expected hourly output × downtime hours.

8. Dashboard

8.1 Indicators (TD §7, ToR §10)

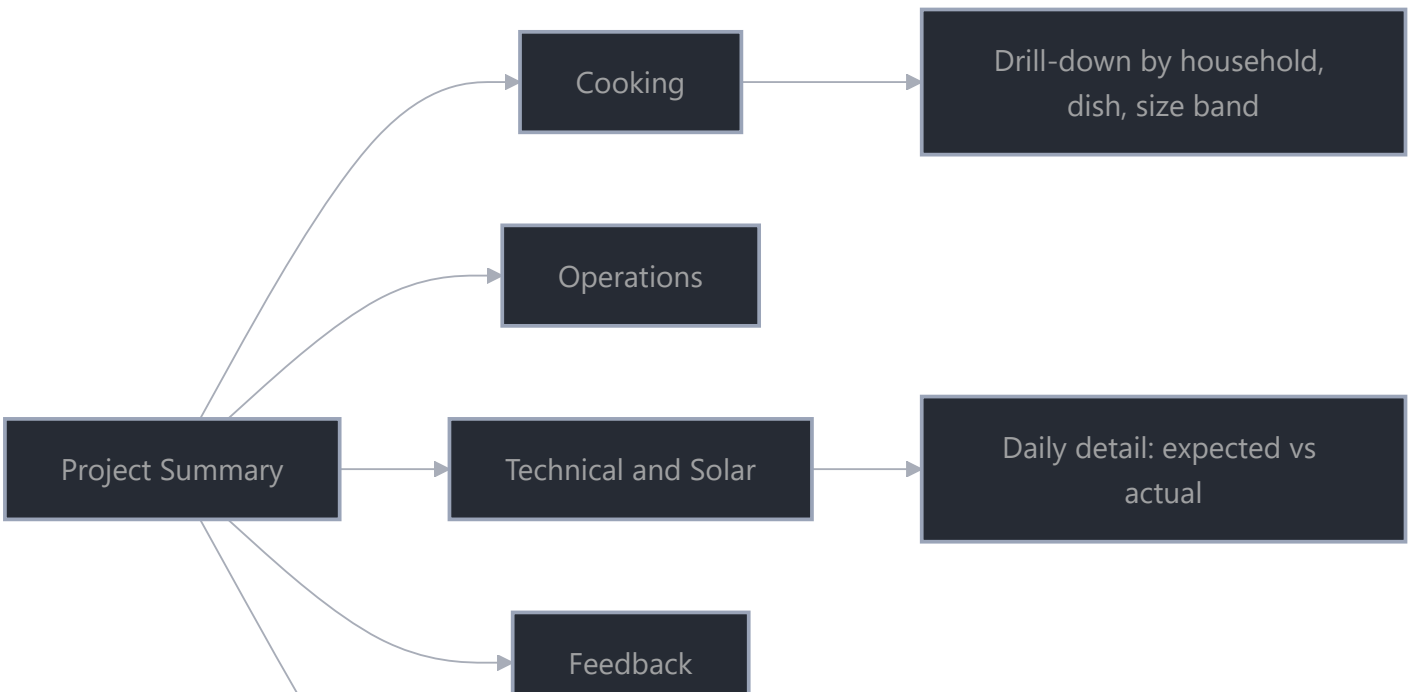
Section	Indicators
A. Overview ✎ (was "Utilities")	Registered households, active households, participation rate, total sessions, completed sessions, session completion rate
B. Cooking Performance	Average cooking time, food type performance, appliance utilization, session-wise performance, incomplete sessions, waiting issues, ★ minutes per person, ★ duration by size band
C. Operational	Waiting issue rate, incomplete session rate, cleaning compliance, technical problems, common user problems, safety incidents, ★ slot utilisation

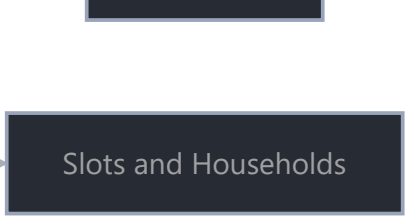
D. Technical	Daily/monthly solar production, morning vs evening SOC, system uptime, technical faults, total downtime, generator running hours, fuel consumption, ★ expected vs actual, ★ efficiency, ★ unused energy
E. Household Feedback	Backup fuel use and expenditure, safety score, privacy score, waiting experience, overall satisfaction, kitchen cleanliness, willingness to continue

8.2 Charts (TD §9 + ToR §11)

#	Chart
1	Cooking sessions: daily / weekly / monthly
2	Solar production trend
3	Battery SOC trend
4	Food type distribution
5	Appliance utilization
6	Technical fault & uptime trend
7	Waiting issue trend
8	Cleaning compliance
9	Backup fuel use trend (+ ★ baseline vs current)
10	User satisfaction trend
11	Safety incident summary
12 ★	Expected vs actual solar
13 ★	Energy flow: consumed / stored / unused
14 ★	Duration by family size band

Filters [ToR §12]: date/period, household, session, indicator, food/appliance category, ★ kitchen, ★ size band. Viewers see coded IDs only.





9. Reports and Export

Report	Contents
Daily	Cooking, solar (★ expected vs actual), battery, faults, downtime, safety, ★ generator
Monthly	Household participation, cooking performance, ★ size-band analysis, technical performance, generator operation, user feedback, backup-fuel use, ★ slot fairness

Export: **Excel, CSV and PDF** ✎. (ToR: Excel/CSV minimum; TD adds PDF). Clean headings, logical formatting, coded IDs.

10. Validation, Security and Privacy

Area	Requirements
Validation [TD §10]	Mandatory fields, household ID validation, date/time validation, duplicate entry detection, automatic calculations, numeric limits, predefined categories
Security	Role-based access, secure login and password hashing, audit trail, regular data backup, controlled data export, secure DB access, restricted admin
Privacy [ToR §16-17]	No unnecessary personal information on public or donor screens, coded IDs, data confidential and owned by the commissioning organisation

11. Corrections Log (ToR vs TD vs this design)

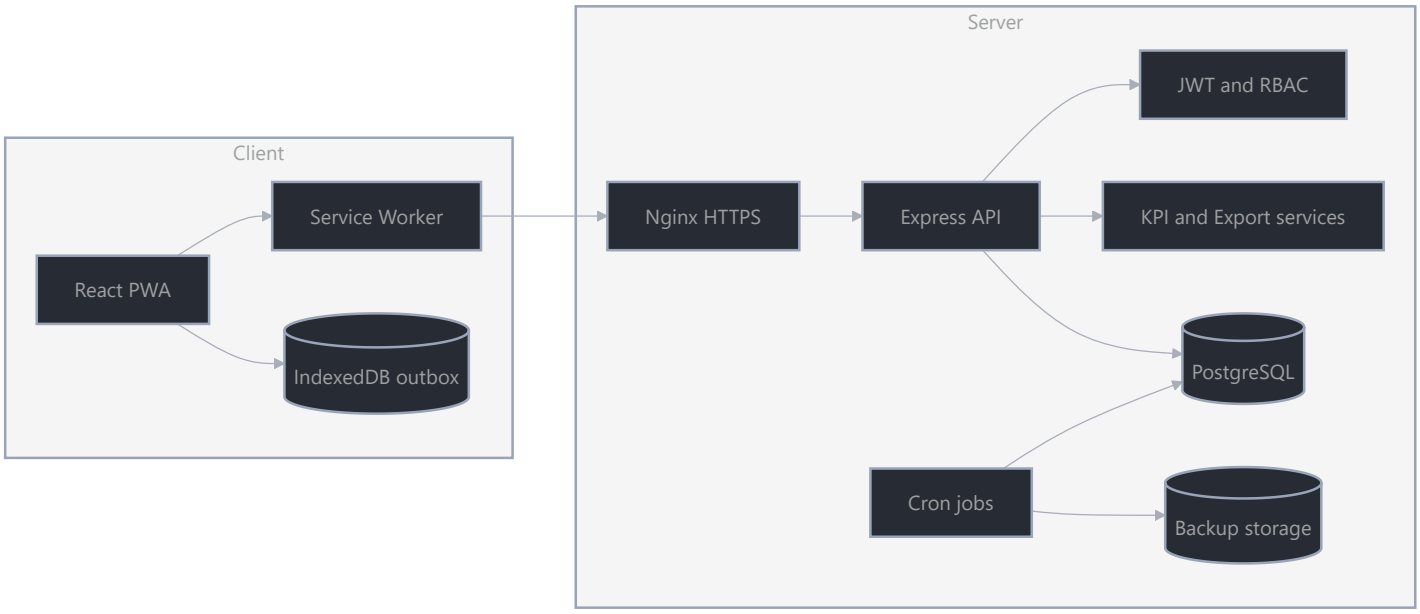
#	Issue	Resolution
1	3 modules (ToR) vs 4 forms / 9 modules (TD)	6 data entities + KPI, dashboard, report layers
2	"Household ID(s)" vs single ID	One session row = one household
3	Export formats differ	Excel + CSV + PDF
4	Dashboard section "Utilities"	Renamed Overview
5	Department/branch-wise users	Kitchen-wise scoping
6	Active household undefined	≥1 attended session in period
7	Scheduled time undefined	Config value per kitchen
8	Fault rate vs fault count	Both kept
9	"Common User Problems" with no data source	Problem category dropdown
10	Safety incident Yes/No only	Severity + note when Yes
11	Baseline fuel and perceived reduction missing	Baseline on household, new survey item
12	Fuel Feed / Re-Fuel Feed ambiguous	Start / refuel / end litres

13	Single food type	Multi-select
14	Average cooking time basis	Completed sessions only
15	Section 29 missing in ToR	Typographical, ignored

12. Added Features Summary ★

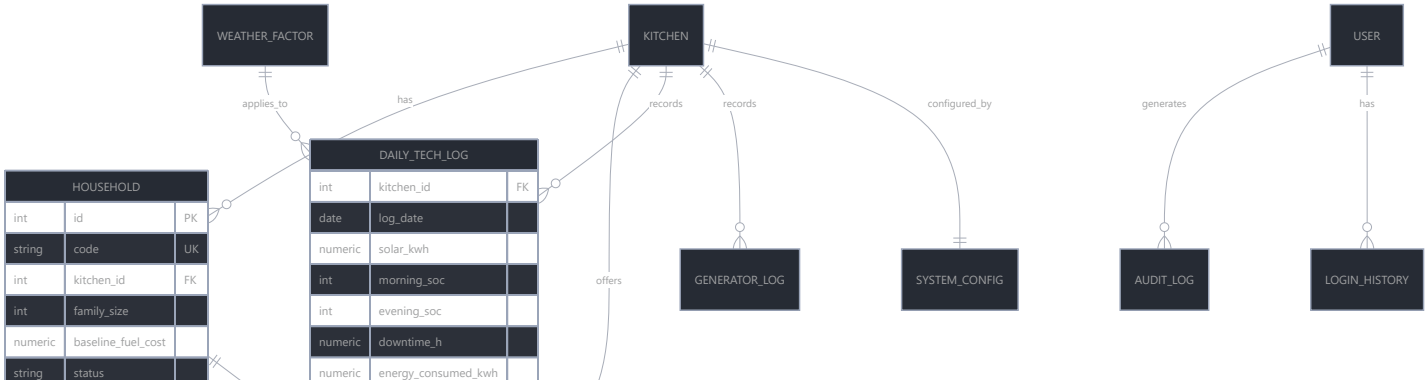
Household Register with family size and fuel baseline; slot scheduling with fair-rotation suggestion and generator flag; persons-cooked-for per session; power source per session; reason categories for waiting and technical problems; size-band analysis in reports and feedback; expected vs actual solar, efficiency and waste; generator fuel consumption; multi-dish sessions; offline sync; kitchen scoping for future locations.

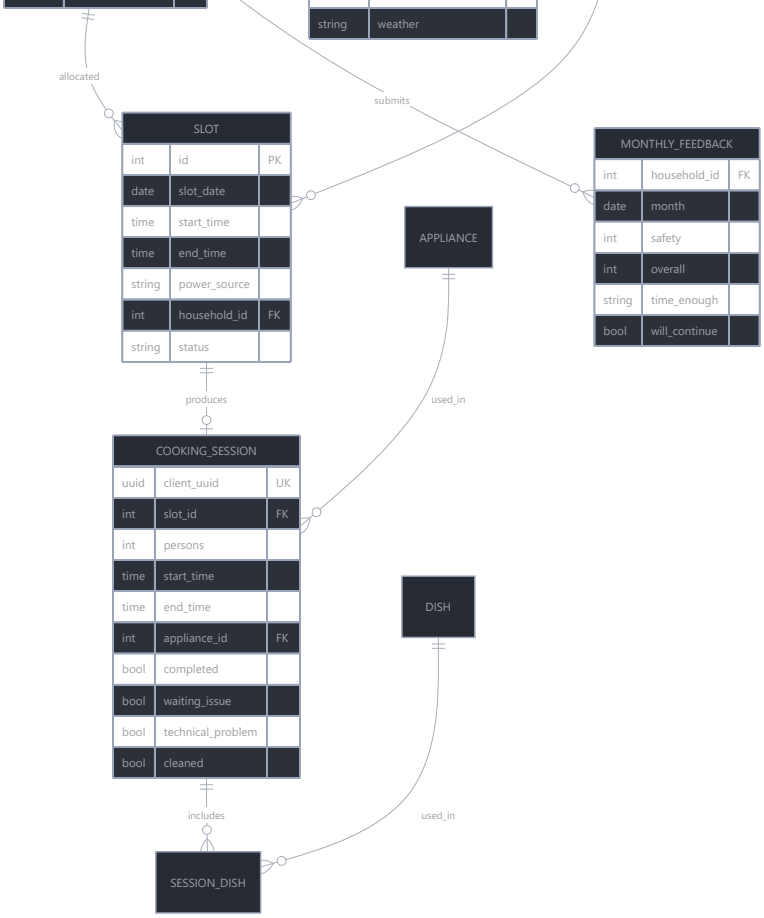
13. Architecture and Tech Stack



Layer	Technology
Database	PostgreSQL with CHECK, UNIQUE, FK constraints and SQL views for KPIs
Backend	Node.js, Express, Zod validation, Prisma or Knex, JWT with refresh, bcrypt, Helmet, rate limiting, pino
Frontend	React (Vite), React Router, TanStack Query, React Hook Form, Recharts
Offline	PWA via Workbox, Dexie (IndexedDB)
Export	exceljs, fast-csv, pdfkit or Puppeteer
Operations	Docker, Nginx, node-cron, pg_dump backups, cloud VPS, domain and HTTPS

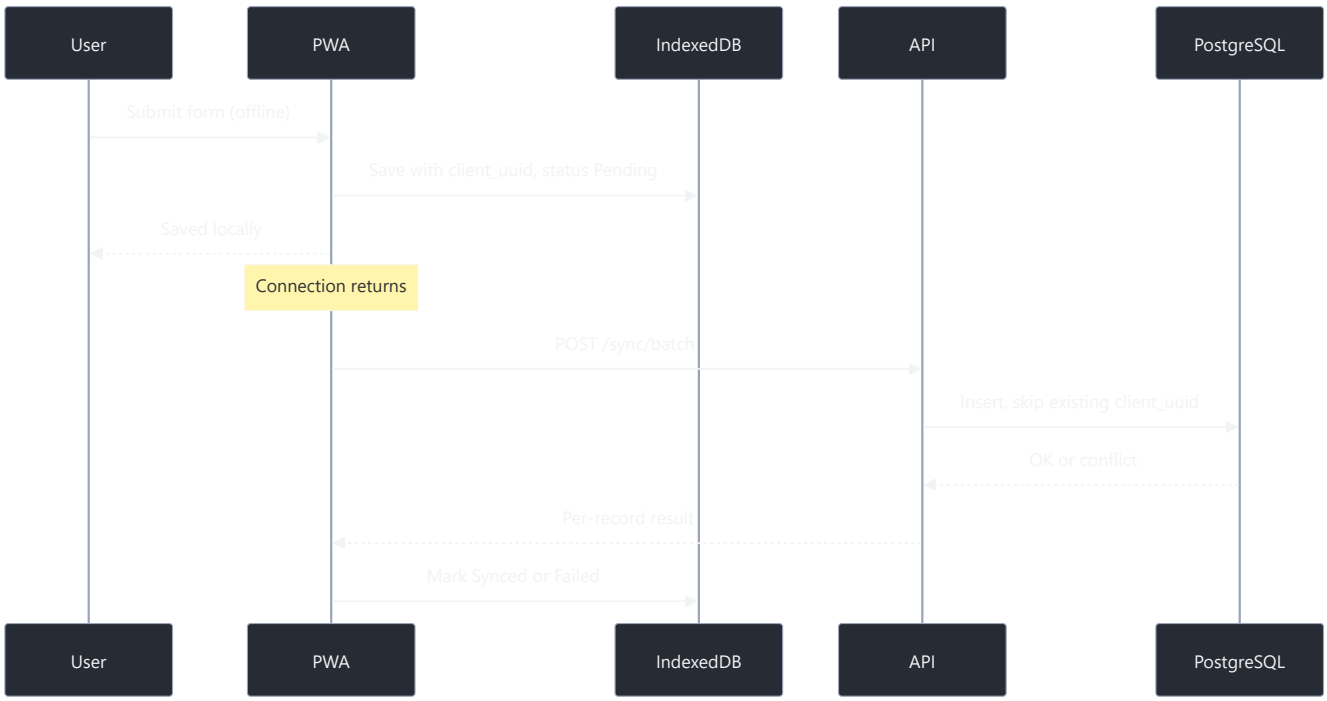
14. ER Diagram





Also: `users` (role, status, kitchen), lookup tables (`problem_category`, `maintenance_type`), `system_config` (kWp, battery kWh, PR, PSH, scheduled hours). Constraints: `UNIQUE(kitchen_id, log_date)`, `UNIQUE(household_id, month)`, `UNIQUE(client_uuid)`, `end_time > start_time`, SOC 0-100, ratings 1-5.

15. Offline and Low-Connectivity Design



1. Service worker caches the app shell; reference data (households, today's slots, lookups) cached locally.
2. Every record gets a **client UUID**, so retries never duplicate (idempotent sync).
3. Sync on reconnect, app open and periodically; retry with backoff; show Pending / Synced / Failed.
4. Unique-key conflicts (e.g., two daily logs for one date) return 409 into an admin review list.
5. Drafts autosave; device time and server time both stored, large clock skew flagged.
6. Offline login uses a time-limited cached token; passwords never stored. Dashboard and exports need

connectivity and show "last updated".

7. Fallback: small payloads and compression for low bandwidth.

16. Other ToR Requirements to Carry Forward

Item	Requirement
Hosting	Cloud hosting + domain for 3 years, basic server and DB monitoring and backup
Testing	Form functionality, validation, DB recording, access, KPI calculation, dashboard, filtering, export, mobile and desktop display
Training	Login, data entry, household/session selection, technical entry, monthly survey, dashboard, filters, export, admin basics, short user guide
Support	3 months post-deployment bug fixes and operational help
Handover	Full owner access (domain, hosting, database), system documentation with flowcharts; data owned by commissioning organisation
Change control	New modules, external integrations, hardware integration and major workflow changes treated separately

17. How to Progress the Project

Step	Work	Done when
1. Foundation	Repo, Docker, PostgreSQL schema, migrations, seed lookups, auth, RBAC, audit log	Roles log in; tables enforce constraints
2. Master data	Kitchen config, household register (16), appliances, dishes, weather factors	Admin can manage all reference data
3. Slots	Slot CRUD, rotation suggestion, swap and status tracking	A week's schedule can be generated and edited
4. Forms	Cooking session, daily technical, generator, monthly feedback with validation and auto-calculation	Each form saves correct data on a phone
5. Offline	PWA, outbox, `/sync/batch`, conflict list	Entries made offline sync without duplicates
6. KPIs	SQL views for all KPIs, size-band analysis, solar expected/efficiency/waste	Results match manual calculation on test data
7. Dashboard	KPI cards, 14 charts, filters, drill-down, coded IDs for viewers	All indicators render and filter correctly
8. Reports	Daily and monthly reports, Excel/CSV/PDF export	Exports open cleanly with proper headings
9. Hardening	Security review, backups, load check, mobile testing, bug fixing	Test checklist (§16) passes
10. Delivery	Deploy, train users, write user guide and system docs, hand over access	Users operate independently

Suggested habit: build with realistic dummy data for 16 households and a month of entries so KPI and chart logic

can be checked early.

18. Open Questions for the Project Team

PV capacity (kWp) and battery capacity (kWh); kitchen operating hours; slots per day and slot length; whether larger families get longer slots; baseline fuel data source; whether energy consumption comes from a meter or is estimated from appliance ratings; final size bands; viewer export rights.